Predicting Earth-Like Similarity in Exoplanets Using Machine Learning

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Abstract

We have analyzed data from NASA's Exoplanet Archive dataset which contains over one million data points and over 6,071 confirmed exoplanets discovered by space- and ground-based missions. Using supervised learning with K-Nearest Neighbor (KNN) classification, we predicted Earth-like similarity in exoplanets and determined their habitability potential by identifying the top 10 Earth-Like exoplanets. Using unsupervised learning clustering algorithms including K Means Clustering, Gaussian Mixture Model (GMM), and Hierarchical Density-Based Spatial Clustering of Applications with Noise (HBDSCAN). we characterized the top 10 Earth-Like exoplanets into planetary categories based on their natural groupings. These natural groupings confirmed that the top 10 Earth-Like exoplanets orbit a Sun-Like star within its Habitable Zone within the Galactic Habitable Zone of the Milky Way Galaxy. By implementing a scalable and reproducible exploratory data analysis, we have characterized exoplanets based on their mass-radius relation. We demonstrated the use of computational modeling in Astronomy and Astrophysics analyses to provide meaningful insights into exoplanet characteristics and accelerate the discovery process of exoplanets.

Introduction

Advances in space missions and computational methods have significantly expanded exoplanet discovery and analysis. Since the launch of NASA's Kepler mission in 2009, followed by K2 and TESS, over 6,701 confirmed exoplanets have been identified, generating large, complex datasets. However, the high dimensionality, variability, and missing data present challenges for characterizing planets and assessing Earth-like similarity and potential habitability. All three telescopes can be referenced in (Figs. 1 – 3).

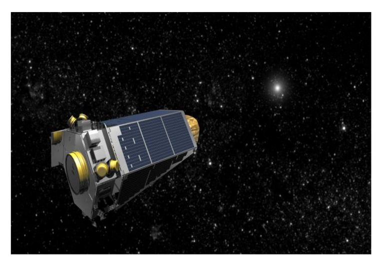

Figure 1: Artist depiction of the Kepler telescope

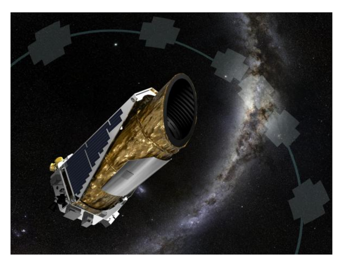

Figure 2: Artist depiction of the K2 telescope

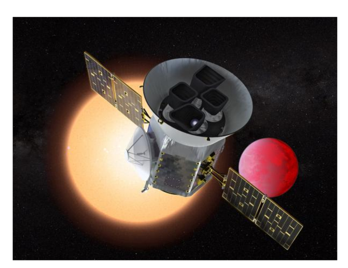

Figure 3: Artist depiction of the TESS telescope

This study applies machine learning and exploratory data analysis (EDA) on the NASA Exoplanet Archive dataset which contains planetary and stellar features of exoplanets and their host stars. This study will predict Earth-Like similarity to identify top Earth-like candidate, evaluate their habitability potential, determine their natural planetary groupings, and classify them based on key planetary and stellar features. Unlike prior work focused primarily on supervised learning, this study integrates both supervised and unsupervised methods for similarity ranking and clustering.

An Earth-Like exoplanet capable of sustaining life is determined by whether it is a rocky Earth-sized or Super-Earth-sized planet orbiting Sun-like (G-type) Main Sequence stars within the star's Habitable Zone inside the Galactic Habitable Zone of the Milky Way Galaxy in accordance with established Astro biological models.

Exoplanet's host stars are also classified by the star types: M, K, G (Sun-like), F, A, B, or O where star types earlier than type F are unlikely to support life due to their short lifespans (Fig. 4). Host stars are also classified by their luminosity class based on the Hertzsprung-Russell diagram. This includes Main Sequence (V) stars, Sub-Giants (IV), Giants (III), Bright Giants (II), Super Giants (I), and White Dwarfs (WD) (Fig. 5).

The star's Habitable Zone depends on the star type (Fig. 6) and describes the optimal distance between the exoplanet and its host star range for liquid water to be sustained which is required for habitability (Fig. 7). Earth-Like similarity depends on a distance within 1 AU. The Galactic Habitable Zone (GHZ) refers to the area within the Milky Way Galaxy which can sustain life due to reduced radiation (Fig. 8).

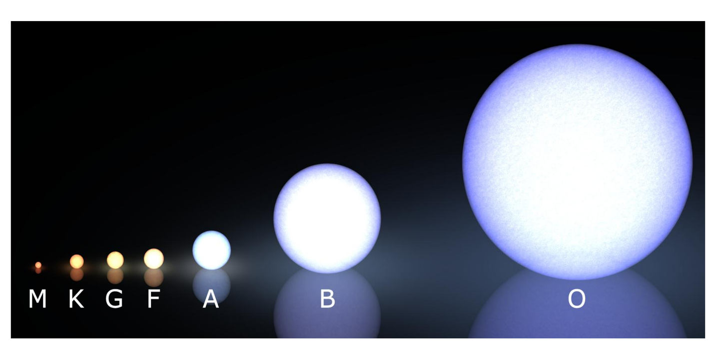

Figure 4: Diagram shows the different star types in order of coldest and longest lifetime to hottest and shortest lifetime

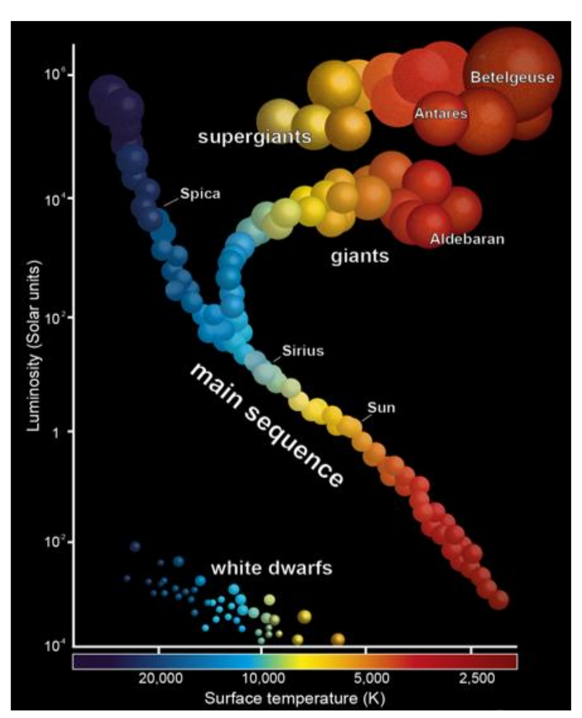

Figure 5: Diagram shows the different luminosity classes and the life stages for stars

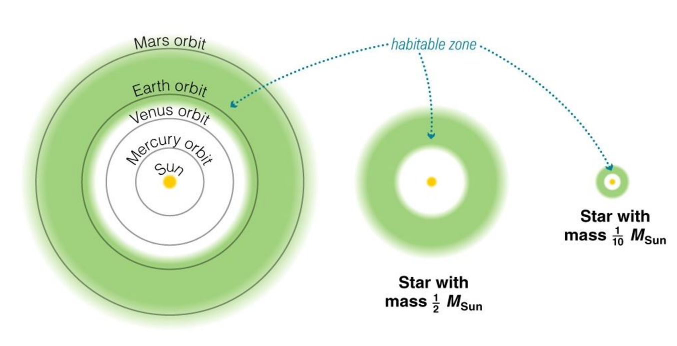

Figure 6: Diagram shows the location of the Habitable Zone which depends on the star type

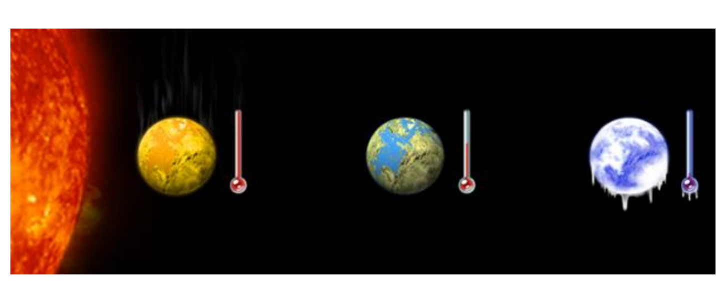

Figure 7: Diagram depicts an artist rendition showing an exoplanet too hot to sustain water, too cold to sustain water, and Earth which holds water.

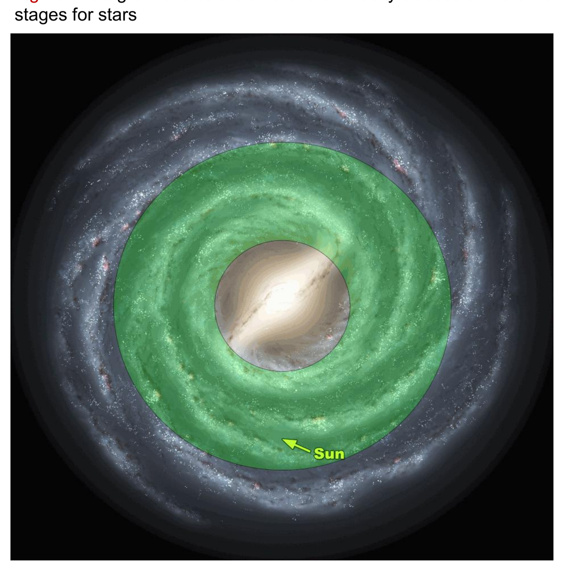

Figure 8: Diagram shows the Galactic Habitable Zone of the Milky Way Galaxy

Objective

- **Supervised machine learning using KNN** → Predefined target row corresponding to Earth used to compare all planetary and stellar features of all exoplanets to Earth and the Sun respectively. Will automatically produce a list of the top 10 Earth-Like exoplanets and insight into their habitability potential.
- **Unsupervised learning using K Means Clustering, GMM, and HDBSCAN** → Predict exoplanet's Earth-Like similarity by identifying their natural planetary groupings without a predefined planetary labels. The results from all three clustering models will be compared to evaluate overall result robustness.
- **Exploratory data analysis** → Explores relationships between stellar and planetary features and classify exoplanets into specific categories through an analysis of their key features.

Materials and Methodology

Software Tools:
- Programming Language: Python
- Development Environment: Jupyter Notebook

Analytical Challenges Present in Data:
- Duplicate rows for a single exoplanet
- Redundant attributes
- Missing data of important attributes
- Typographical errors

Initial Preprocessing Steps:
- Raw dataset: 39,235 rows × 92 columns.
- Removed redundant features and renamed variables → 21 columns retained.
- Removed unconfirmed/controversial planets

Outlier Detection:
- Duplicate rows per exoplanet (multiple authors, varying measurements).
- Grouped data by planet name and applied Z-score–based outlier detection within each group.
- Population mean (μ) and standard deviation (σ) computed excluding missing values.
- If σ ≤ 0.01 → variability negligible → no outliers flagged.
- Otherwise, outliers identified using:

$$z = \frac{x - \mu}{\sigma}$$

- Thresholds:
  - |z| > 3 for most features (99.7% confidence)
  - |z| > 1 for sensitive stellar parameters (mass, radius)
- Outliers replaced with null values (to avoid skew and false zero assumptions).
- Entries with (NASA-verified measurements) were retained.

Handling Duplicate Observations:
- Aggregated duplicates:
  - Numerical features → mean
  - Categorical features → mode
- 6,071 confirmed exoplanets (Jan 2026)

Missing Data and Imputation:
- Missing values for key features imputed using relevant astronomical formulas

Feature Engineering:
- Creation of columns using astronomical formulas
  - Stellar Luminosity
  - Planet Radial Velocity
  - Habitable Zone

Final Preprocessing Steps:
- Removed rows and columns with excessive non-imputable missing data

Final Dataset:
- 5,419 confirmed exoplanets remain
- 16 key features

Principal Component Analysis (PCA):
- High dimensional data (34 features) reduced into 2 features known as Principal Components 1 and 2 for visualization purposes.

K Nearest Neighbor (KNN) Models

- Identify the top 10 most Earth-like exoplanets.
- Each planet compared to synthetic "Earth" reference point in feature space.
- Similarity measured via distance → smaller distance → closer to Earth → more Earth-Like.
- Two KNN models using two distance metrics:
  - **KNN Model #1** - Manhattan distance → sensitive to deviations between a feature
  - **KNN Model #2** - Euclidean distance → considers similarity for all features combined
  - KNN used as a similarity ranking.
- Heavier weights applied to features important for determining Earth-Like similarity and habitability

K-Means Clustering Models

- Identify natural groupings of exoplanets into their planetary systems.
- Model applied to the top 10 Earth-like planets (from KNN Models) to determine their cluster membership.
- Partitions data into k clusters by minimizing within-cluster variance
- Reveals underlying structure in exoplanet populations
- Data preprocessing:
  - One-hot encoding for categorical features and Feature scaling
- Two K-Means models:
  - **Model #1**: Includes outliers, k = 6 clusters
  - **Model #2**: Removes outliers, k = 4 clusters

Gaussian Mixture Models (GMM)

- Applied GMM clustering to capture more flexible, overlapping exoplanet groupings.
- GMM models probabilistic cluster membership.
- Data preprocessing:
  - One-hot encoding (categorical features) and Feature scaling
- Two GMM models:
  - **Model #1**: Includes outliers
  - **Model #2**: Removes outliers

Hierarchical Density-Based Spatial Clustering HDBSCAN

- Applied HDBSCAN to identify clusters of varying density while separating noise/outliers.
- Detects clusters of different shapes and densities
- Automatically identifies noise points

Modeling and EDA Results

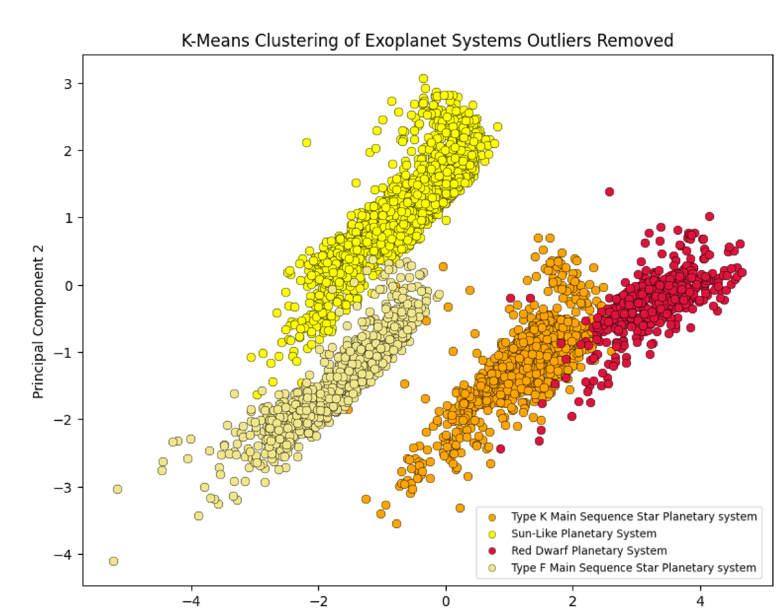

Figure 9: Graph shows the 4 clusters identified by the K-Means Cluster model for all confirmed exoplanets with 535 outlier exoplanets removed due to anomalous features present.

K-Means Clustering Results for Top 5 Earth-Like Exoplanets			
Rank	**Planet**	**Cluster_ID**	**Cluster_Type (Planetary System)**
1	Kepler-452 b	1	Sun-Like Planetary System
2	Kepler-1638 b	1	Sun-Like Planetary System
3	Kepler-51 e	1	Sun-Like Planetary System
4	Kepler-1606 b	1	Sun-Like Planetary System
5	Kepler-1701 b	1	Sun-Like Planetary System

Figure 10: Table shows the K-Means Clustering model correctly predicted the top Earth-Like exoplanets found from the KNN Model #1 and KNN Model #2 belongs to a Sun-Like Planetary System

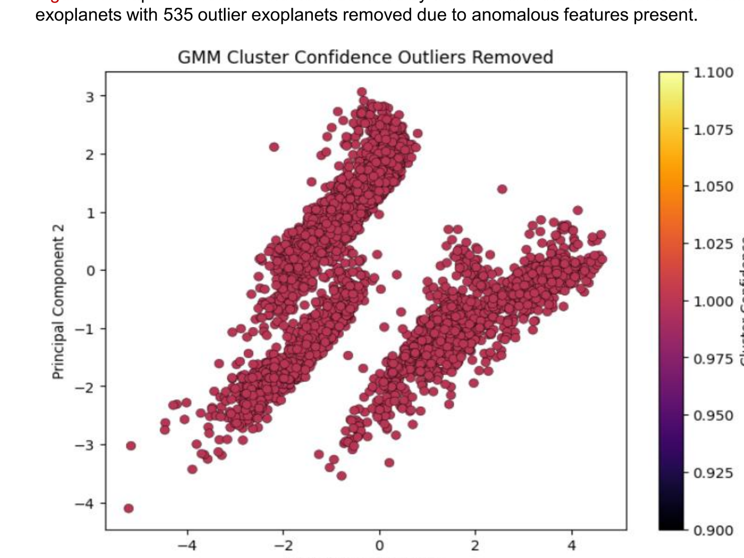

Figure 11: Graph shows the 4 clusters identified by the GMM which are consistent with the K-Means model. This shows the confidence of the model in assigning all exoplanets to their appropriate clusters. Most planets have >99% probability of belonging to the correct cluster.

GMM Results for Top 5 Earth-Like Exoplanets			
Rank	**Planet**	**Cluster_ID**	**Cluster_Type (Planetary System)**
1	Kepler-452 b	1	Sun-like stars hosting Super-Earth-sized rocky planets
2	Kepler-1638 b	1	Sun-like stars hosting Super-Earth-sized rocky planets
3	Kepler-51 e	1	Sun-like stars hosting Super-Earth-sized rocky planets
4	Kepler-1606 b	1	Sun-like stars hosting Super-Earth-sized rocky planets
5	Kepler-1701 b	1	Sun-like stars hosting Super-Earth-sized rocky planets

Figure 12: Table shows the GMM correctly predicted the top Earth-Like exoplanets found from the KNN Model #1 and KNN Model #2 belongs to a Planetary System where Super-Earth-sized exoplanets orbit a Sun-Like star

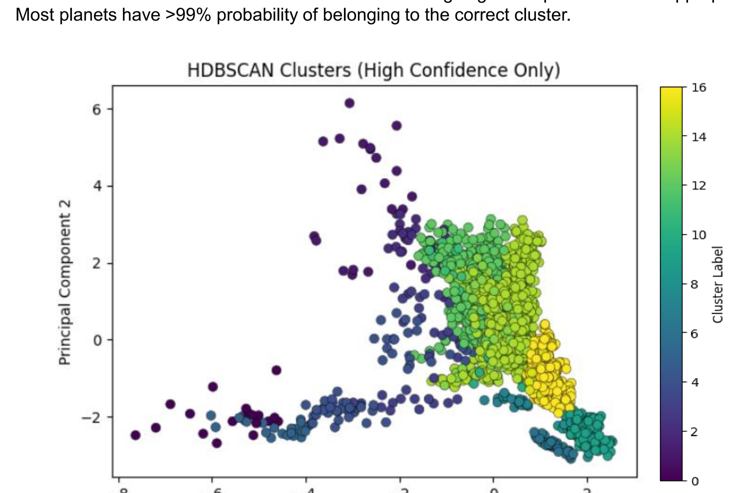

Figure 13: Graph shows 16 clusters identified by the HDBSCAN model for all exoplanets where outlier exoplanets and exoplanets with membership probability < 80% automatically removed.

HDBSCAN Results for Top 5 Earth-Like Exoplanets				
Rank	**Planet**	**Cluster_ID**	**Cluster_Type (Planetary System)**	**Cluster Probability**
1	Kepler-452 b	7	Sun-like stars hosting Super-Earth-sized rocky planets in the habitable zone	100%
2	Kepler-1638 b	7	Sun-like stars hosting Super-Earth-sized rocky planets in the habitable zone	85.75%
3	Kepler-51 e	7	Sun-like stars hosting Super-Earth-sized rocky planets in the habitable zone	100%
4	Kepler-1606 b	7	Sun-like stars hosting Super-Earth-sized rocky planets in the habitable zone	100%
5	Kepler-1701 b	7	Sun-like stars hosting Super-Earth-sized rocky planets in the habitable zone	100%

Figure 14: Table shows the GMM correctly predicted the top Earth-Like exoplanets found from the KNN Model #1 and KNN Model #2 belongs to a Planetary System where Super-Earth-sized exoplanets orbit a Sun-Like star in its Habitable Zone with the associated membership probabilities

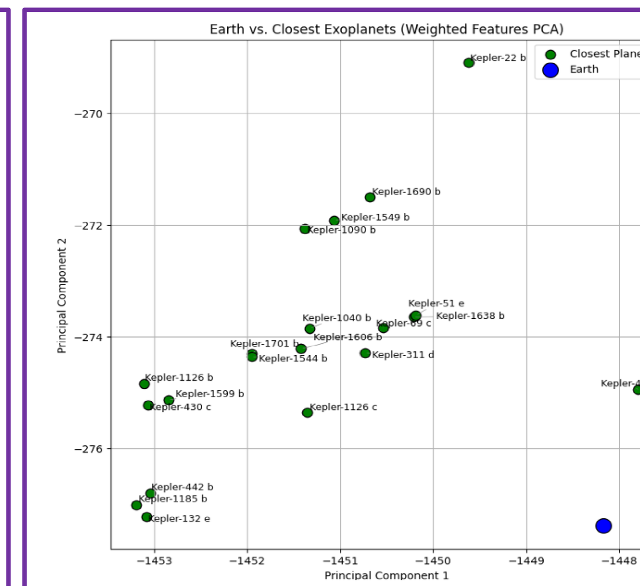

Figure 15: Graph which depicts the top 10 Earth-Like exoplanets (shown in green) in proximity to Earth (shown in blue).

Top 10 Earth-Like Exoplanets KNN Model Comparison				
Rank	**KNN Model #1**	**Manhattan Distance**	**KNN Model #2**	**Euclidean Distance**
1	Kepler-452 b	10.1463	Kepler-452 b	5.2408
2	Kepler-1638 b	16.6309	Kepler-1638 b	7.1452
3	Kepler-51 e	16.7816	Kepler-51 e	7.7652
4	Kepler-1606 b	19.284	Kepler-1606 b	8.2007
5	Kepler-1701 b	20.1409	Kepler-1701 b	8.466
6	Kepler-22 b	20.8885	Kepler-1126 c	9.4643
7	Kepler-1126 c	22.4542	Kepler-69 c	9.9627
8	Kepler-1549 b	22.9228	Kepler-1185 b	10.35
9	Kepler-1690 b	23.2328	Kepler-442 b	10.3699
10	Kepler-69 c	23.5054	Kepler-311 d	10.3847

Figure 16: Table shows the top 10 Earth-Like exoplanets found from the KNN Model #1 (Manhattan Distance) and KNN Model #2 (Euclidean Distance). The top 5 Earth-Like exoplanets remain consistent showing robustness.

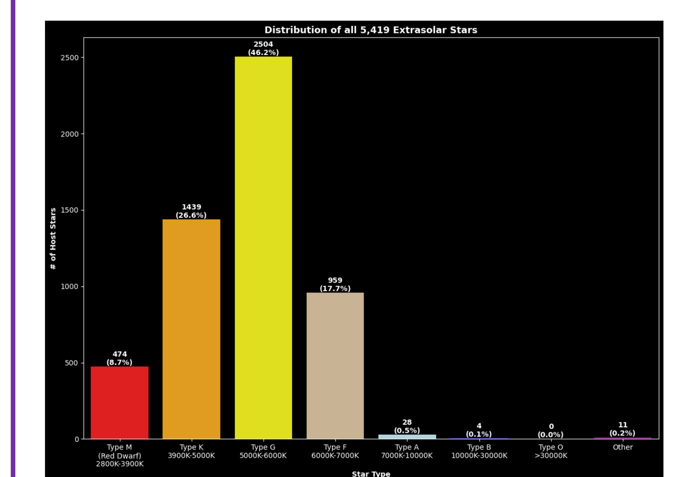

Figure 17: Bar Chat of the Distribution of all 5,419 Extrasolar Stars Characterized by spectral type

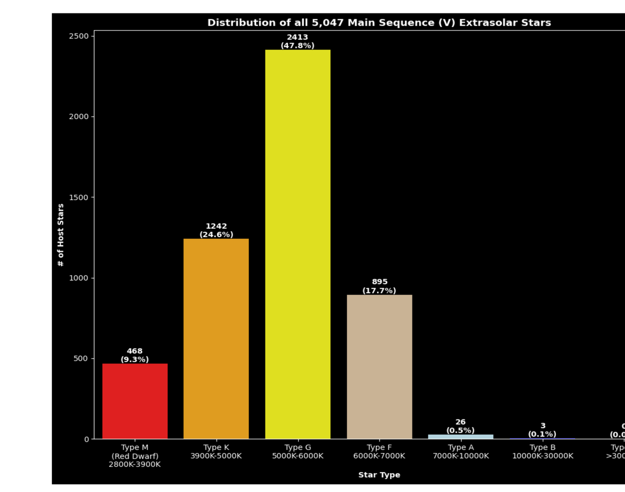

Figure 18: Bar Chat of the Distribution of all 5,047 Main Sequence Extrasolar Stars Characterized by spectral type

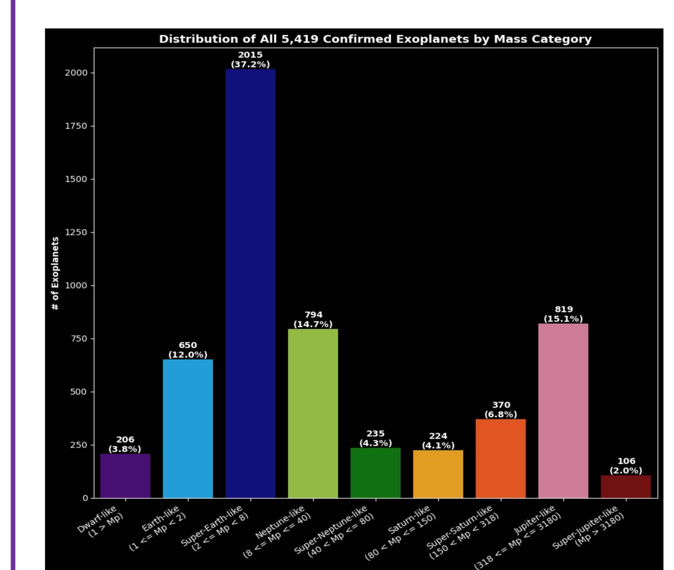

Figure 19: Bar Chart of Mass Characterization for all 5,419 Exoplanets which orbits a Main Sequence (V), Sub Giant (IV), Red Giant (III) Host Star.

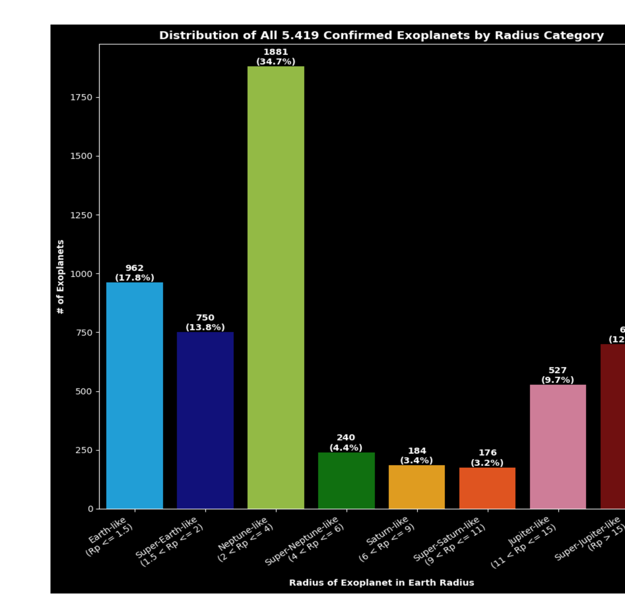

Figure 20: Bar Chart of Radius Characterization for all 5,419 Exoplanets which orbits a Main Sequence (V), Sub Giant (IV), Red Giant (III) Host Star.

Conclusion and Future Work

KNN Model #1 (Manhattan Distance) and KNN Model #2 (Euclidean Distance) shows:
- The **most Earth-Like exoplanet** is **Kepler-452 b.** Followed by **Kepler-1638 b, Kepler-51 e, Kepler-1606 b,** and **Kepler-1701 b** respectively as shown in Fig. 15.
- **Top 5 Earth-like exoplanets remain consistent** using both Manhattan and Euclidean distance metrics. So, **top-ranked Earth-Like planets are robust.**
- **Mid-ranked candidates are sensitive to distance metric choice.** This suggests that **similarity rankings less stable beyond the top candidates.**
- The Fig. 16 supports these findings due to close clustering between top Earth-Like planets and Earth.

K-Means Clustering, GMM, and HDBSCAN shows agreement:
- **Consistent results** for the planetary groupings of the top 10 Earth-Like exoplanets found **by both KNN models** which solidifies validity of results.
- **All 3 clustering models** successfully **identified the natural planetary groupings of all confirmed exoplanets** (Figs. 9, 11, 13).
- Groupings show **top 5 exoplanets** are **likely rocky Earth-sized or Super-Earth-sized planets orbiting a Type-G (Sun-Like) star within the Habitable Zone of its star within the Galactic Habitable Zone.** Thus, making the **top 5 Earth-Like planets potentially habitable.** (Figs. 10, 12, 14)

K-Means Clustering, GMM, and HDBSCAN comparison:
- **K-Means classified planets** based on **their stellar features. GMM classified planets** based on **stellar and planetary features. HDBSCAN classified planets** based on **all key features.**
- **GMM's high membership probability overall** indicates model was **very confident** in correctly **assigning all exoplanets (including top 10 Earth-Like exoplanets) to clusters. HDBSCAN's high membership probability for each cluster** demonstrated **high probabilities between 85% to 100% confidence** in correctly **assigning top 10 Earth-Like exoplanets to clusters.**
- **Clustering structure for K-Means and GMM same** due to manually removing outlier planets. **Clustering structure for HDBSCAN different** since outlier planet automatically removed by model.

EDA results show Earth and associated exoplanets in area rich with Sun-Like Stars:
- **Majority of extrasolar stars** are **Type-G (Sun-Like) stars.** When looking at just **Main Sequence (V) stars**, **a majority** are **Type-G (Sun-Like) stars** (Figs. 17 - 18).
- **Majority of exoplanets Super-Earth-Like in radius, and Neptune-Like in mass** (Figs. 19 - 20).

Future Work:
- Will explore the use of additional unsupervised learning models to predict habitability potential for all exoplanets and rank exoplanets by their habitability potential.

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